Dynatrace OTLP Integration

> **Series:** OTEL | **Notebook:** 7 of 8 | **Created:** January 2026 | **Last Updated:** 01/30/2026

Complete Setup for OpenTelemetry with Dynatrace

This notebook provides end-to-end configuration for sending OpenTelemetry data to Dynatrace, including authentication, endpoints, and verification.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS or Managed with OTLP enabled
Permissions	Token creation access
Knowledge	OTEL-01 through OTEL-06

1. Dynatrace OTLP Endpoints

SaaS Endpoints

Signal	gRPC Endpoint	HTTP Endpoint
All	`{env-id}.live.dynatrace.com:443`	`https://{env-id}.live.dynatrace.com/api/v2/otlp`
Traces	Same	`https://{env-id}.live.dynatrace.com/api/v2/otlp/v1/traces`
Metrics	Same	`https://{env-id}.live.dynatrace.com/api/v2/otlp/v1/metrics`
Logs	Same	`https://{env-id}.live.dynatrace.com/api/v2/otlp/v1/logs`

ActiveGate Endpoints

For on-premises or network-restricted environments:

Signal	Endpoint
All	`https://{activegate-host}:9999/e/{env-id}/api/v2/otlp`

Environment ID

Find your environment ID:
1. Log into Dynatrace
2. Look at the URL: `https://abc12345.live.dynatrace.com`
3. `abc12345` is your environment ID

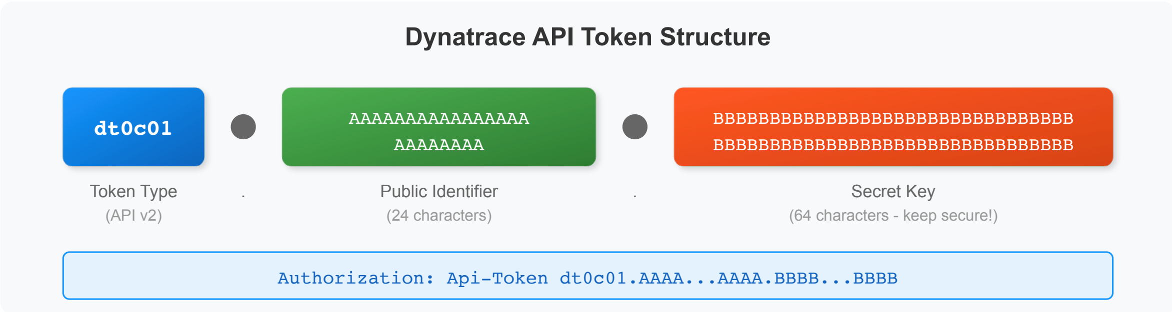
2. Authentication Setup

Create API Token

1. Navigate to **Settings > Access tokens**
2. Click **Generate new token**
3. Name: `otel-ingest-token`
4. Select scopes:

Scope	Signal	Required
`openTelemetryTrace.ingest`	Traces	Yes (for traces)
`metrics.ingest`	Metrics	Yes (for metrics)
`logs.ingest`	Logs	Yes (for logs)
`events.ingest`	Events	Optional

Token Format



Header Format

Use the `Api-Token` prefix in the Authorization header:

```
Authorization: Api-Token &lt;your-full-token&gt;
```

3. Collector Configuration

Image Versioning Best Practice

> ⚠️ **Important:** Always pin your OTel Collector image to a specific version. Using `latest` can cause unexpected behavior during upgrades.

```
# ❌ Avoid - Non-deterministic deployments
image: otel/opentelemetry-collector-contrib:latest

# ✅ Recommended - Pin to specific version
image: otel/opentelemetry-collector-contrib:0.96.0
```

Check the OpenTelemetry Collector releases for the current stable version.

Complete Collector Config for Dynatrace

```yaml
# otel-collector-config.yaml
receivers:
  otlp:
    protocols:
      grpc:
        endpoint: 0.0.0.0:4317
      http:
        endpoint: 0.0.0.0:4318

processors:
  batch:
    timeout: 10s
    send_batch_size: 1000

  memory_limiter:
    check_interval: 1s
    limit_mib: 800
    spike_limit_mib: 200

  # Add service.name if missing
  resource:
    attributes:
      - key: service.name
        value: "unknown-service"
        action: insert

exporters:
  otlphttp/dynatrace:
    endpoint: https://${DT_ENV_ID}.live.dynatrace.com/api/v2/otlp
    headers:
      Authorization: Api-Token ${DT_API_TOKEN}
    retry_on_failure:
      enabled: true
      initial_interval: 5s
      max_interval: 30s
      max_elapsed_time: 300s

  debug:
    verbosity: detailed

extensions:
  health_check:
    endpoint: 0.0.0.0:13133

service:
  extensions: [health_check]
  pipelines:
    traces:
```

```
      receivers: [otlp]
      processors: [memory_limiter, resource, batch]
      exporters: [otlphttp/dynatrace]
    metrics:
      receivers: [otlp]
      processors: [memory_limiter, resource, batch]
      exporters: [otlphttp/dynatrace]
    logs:
      receivers: [otlp]
      processors: [memory_limiter, resource, batch]
      exporters: [otlphttp/dynatrace]
```

Resource Sizing Guide

Environment	CPU Limit	Memory Limit	Batch Size
Dev/Test	200m	256Mi	500
Staging	500m	512Mi	1000
Production	1000m	1Gi	2000

Running with Environment Variables

```
export DT_ENV_ID="abc12345"
export DT_API_TOKEN="&lt;your-api-token&gt;"

otelcol-contrib --config otel-collector-config.yaml
```

4. Direct SDK Export

Python Direct to Dynatrace

```python
import os
from opentelemetry import trace
from opentelemetry.sdk.trace import TracerProvider
from opentelemetry.sdk.trace.export import BatchSpanProcessor
from opentelemetry.exporter.otlp.proto.http.trace_exporter import OTLPSpanExporter
from opentelemetry.sdk.resources import Resource

# Configuration
DT_ENV_ID = os.environ["DT_ENV_ID"]
DT_API_TOKEN = os.environ["DT_API_TOKEN"]

# Resource with service name
resource = Resource.create({
    "service.name": "my-python-app",
    "service.version": "1.0.0",
    "deployment.environment": "production"
})

# Setup tracer
provider = TracerProvider(resource=resource)
exporter = OTLPSpanExporter(
    endpoint=f"https://{DT_ENV_ID}.live.dynatrace.com/api/v2/otlp/v1/traces",
    headers={"Authorization": f"Api-Token {DT_API_TOKEN}"}
)
provider.add_span_processor(BatchSpanProcessor(exporter))
trace.set_tracer_provider(provider)
```

Java with System Properties

```bash
# Set environment variables first
export DT_ENV_ID="abc12345"
export DT_API_TOKEN="&lt;your-api-token&gt;"

# URL-encode the token for the header (space becomes %20)
java -javaagent:opentelemetry-javaagent.jar \
  -Dotel.service.name=my-java-app \
  -Dotel.exporter.otlp.endpoint=https://${DT_ENV_ID}.live.dynatrace.com/api/v2/otlp
\
  -Dotel.exporter.otlp.headers="Authorization=Api-Token%20${DT_API_TOKEN}" \
  -jar app.jar
```

Node.js

```
const { NodeSDK } = require('@opentelemetry/sdk-node');
const { OTLPTraceExporter } = require('@opentelemetry/exporter-trace-otlp-http');
const { Resource } = require('@opentelemetry/resources');

const sdk = new NodeSDK({
  resource: new Resource({
    'service.name': 'my-node-app',
  }),
  traceExporter: new OTLPTraceExporter({
    url:
`https://${process.env.DT_ENV_ID}.live.dynatrace.com/api/v2/otlp/v1/traces`,
    headers: {
      'Authorization': `Api-Token ${process.env.DT_API_TOKEN}`,
    },
  }),
});

sdk.start();
```

5. Entity Mapping

How Dynatrace Maps OTel Data

OTel Resource Attribute	Dynatrace Entity	Notes
`service.name`	Service	Required for service detection
`service.namespace`	Service group	Optional grouping
`host.name`	Host	Links to host entity
`k8s.namespace.name`	K8s namespace	Links to K8s entities
`k8s.pod.name`	Process group	Links to pod

Essential Resource Attributes

Always set these for proper entity mapping:

```
resource = Resource.create({
    # Required
    "service.name": "checkout-api",

    # Recommended
    "service.version": "1.2.3",
    "service.namespace": "ecommerce",
    "deployment.environment": "production",

    # For K8s
    "k8s.namespace.name": "checkout",
    "k8s.pod.name": "checkout-api-abc123",
    "k8s.deployment.name": "checkout-api",
})
```

6. Span Attributes for Dynatrace

Dynatrace-Specific Attributes

Attribute	Purpose	Example
`dt.entity.process_group_instance`	Link to PGI	`PROCESS_GROUP_INSTANCE-XXX`
`dt.span.type`	Override span type	`DATABASE`, `MESSAGING`
`dt.source`	Data source	`opentelemetry`

Semantic Conventions Dynatrace Uses

Convention	Dynatrace Feature
`http.*`	HTTP service detection
`db.*`	Database call analysis
`messaging.*`	Message queue tracking
`rpc.*`	RPC call tracking

Example with Rich Attributes

```
with tracer.start_as_current_span("checkout") as span:
    # Standard semantic conventions
    span.set_attribute("http.method", "POST")
    span.set_attribute("http.url", "https://api.example.com/checkout")
    span.set_attribute("http.status_code", 200)
    span.set_attribute("http.route", "/checkout")

    # Business context
    span.set_attribute("order.id", order_id)
    span.set_attribute("order.total", order_total)
```

```
// Verify OTel traces are arriving
fetch spans, from:-1h
| filter isNotNull(otel.library.name)
| summarize count = count(), by:{service.name, otel.library.name}
| sort count desc
| limit 20
```

```
// Check recent OTel spans
fetch spans, from: now() - 1h
| filter isNotNull(otel.library.name)
| fields timestamp, service.name, span.name, duration
| sort timestamp desc
| limit 30
```

```
// View OTel data by instrumentation scope
fetch spans, from:-1h
| filter isNotNull(otel.scope.name)
| summarize count = count(), by:{otel.scope.name}
| sort count desc
| limit 10
```

7. Verification

Verify Data in Dynatrace

1. **Services**: Go to **Services** - OTel services appear with OTel icon
2. **Traces**: Go to **Distributed Traces** - Filter by service
3. **Metrics**: Go to **Metrics** - Search for custom metric names
4. **Logs**: Go to **Logs & Events** - Filter by trace_id

Troubleshooting Checklist

Check	How
Token scopes	Settings > Access tokens
Collector logs	`otelcol-contrib --config config.yaml`
Network connectivity	`curl -v https://{env}.live.dynatrace.com/api/v2/otlp`
service.name set	Check resource attributes

Common Issues

Issue	Cause	Fix
401 Unauthorized	Invalid token	Check token, regenerate
403 Forbidden	Missing scope	Add required scopes
No data	service.name missing	Add resource attribute
Partial data	Network timeout	Check batch size, retry config

8. Hybrid with OneAgent

Using OTel + OneAgent

Scenario	Recommendation
OneAgent-supported tech	Let OneAgent auto-instrument
Custom spans needed	Add OTel manual spans
Serverless	Use OTel (no agent)
Unsupported language	Use OTel SDK

Trace Context Propagation

OneAgent and OTel both support W3C Trace Context, enabling unified traces:

```
Service A (OneAgent) → Service B (OTel) → Service C (OneAgent)
                 ↓                  ↓
            Dynatrace traces show complete path
```

Configuration for Hybrid

Ensure both use W3C propagation:

```
# Python OTel
from opentelemetry.propagate import set_global_textmap
from opentelemetry.propagators.composite import CompositePropagator
from opentelemetry.propagators.w3c.traceparent import W3CTraceparentPropagator
from opentelemetry.propagators.w3c.tracestate import W3CTracestatePropagator

set_global_textmap(CompositePropagator([
    W3CTraceparentPropagator(),
    W3CTracestatePropagator()
]))
```

Summary

In this notebook, you learned:

- Dynatrace OTLP endpoints for SaaS and ActiveGate
- API token creation with required scopes
- Complete Collector configuration
- Direct SDK export configuration
- Entity mapping and resource attributes
- Span attributes that enhance Dynatrace analysis
- Verification and troubleshooting
- Hybrid deployment with OneAgent

References

- Dynatrace OpenTelemetry
- OTLP Ingest API
- Entity Mapping

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.